Centennial Jing–Zhang AI Innovation Belt Urban Design

Jing–Zhang Smart Vein · Unbounded Green

v9.5 (Formal Ready) | 2026-08-12 | YuanYii

With Jing–Zhang Heritage Park as the cultural spine, build a three–tier spatial control system (study—design—key areas), forming "one belt, three cores, ten districts linked, blue–green slow–mobility ring"; Block Token zone licensing is the core of the urban AI governance protocol.

11.41 km^2 Overall Design Scope	31.1% Green Ratio	25.3% Public Space Ratio	TBD Building Footprint

TBD Building Density	3 areas · 368.4 ha Key Areas	85.6% Green 300m Coverage	20.6% TOD 500m Union

Sheets generated on provisional constraints; do not constitute statutory regulatory red lines. All geometry, metrics and data are conceptual and must be re-verified by professional departments after official red lines are released.

Contents

3 01 Design Basis & Source List
Official package takes announcement & taskbook as primary authority; all sources registered item-by-item in sources.json

Source Grading

Sources graded in four authority levels: official_public (brief/announcement/agent taskbook) as primary authority; repository_public (public registry & agent-readable navigation) as process material; provisional for temporary boundaries; background for think-tank material. The official package follows official_public.

Basis System

Source ID	Use	Authority
PUBLIC-BRIEF	Public brief (primary authority)	official_public
OFFICIAL-ANNOUNCEMENT	Pre-qualification announcement 1.3/1.4/1.5	official_public
AGENT-TASKBOOK	Agent taskbook (6 tasks / 13-dim review)	official_public
SITE-PACKAGE	Design brief / enumeration / permitted spaces / metrics	official_public
SOURCE-REGISTRY	Public source registry (authority/licence/use)	repository_public
PROCESSED-FACT-PACK	Agent-readable navigation layer	repository_public
BOUNDARY-SOURCE	Provisional boundary (temporary, coarse)	provisional
KEY-AREA-SOURCE	Provisional key areas	provisional
PUBLIC-THINKTANK-REGISTRY	Public think-tank registry	background

4 02 Current-State Review & Data Boundary

Data gaps and boundaries declared honestly; all claims rollback-able and verifiable

Data Boundary Statement

Sheets generated on provisional constraints; do not constitute statutory regulatory red lines. All geometry, metrics and data are conceptual and must be re-verified by professional departments after official red lines are released.

Data gaps & boundary

- Official red lines & key-area polygons not released — all boundaries/areas are provisional concepts (official_boundary=false)
- Building census (year/structure/legality) not released — buildings.geojson all status=unknown; demolish/convert/retain is a concept draft
- Regulatory FAR & total floor area not released — floor_area_ratio=unknown, pending official data
- Real road/rail/station data not released — transport metrics are synthetic scenario estimates, pending on-site verification

District	Action	Targets	Basis
Origin Community	Preserve & Restore	Qinghuayuan station site, university research buildings	Heritage status, build year (public data)
Zhongzhiyuan	Retrofit & Upgrade	Qinghe-frontage legacy plants & low-efficiency buildings	Industrial transition needs (conceptual)
Dazhongsi	Demolition Study	Illegal structures & hazardous temporary buildings	Safety hazards (conceptual, pending engineering survey)
full line	New Build Reserve	Edge stations, TOD corridors & other small facilities	Scenario demand (conceptual)

Conceptual demolish/convert/retain scenario draft (confidence=low, pending official building census; not an engineering or demolition arrangement)

5 03 Three-Tier Scope Framework

Three tiers: study — overall design — key areas

Three-Tier Scope Framework

Three tiers in sequence: study tier (43.6 km^2) for positioning & regional synergy; design tier (11.41 km^2) for land-use, blue-green, transport & urban-form framework; key-areas tier (368.4 ha) for refined design of three core districts. No overlap, no gaps; metrics traceable tier by tier.

Overall Concept

With Jing-Zhang Heritage Park as the cultural spine, build a three-tier spatial control system (study—design—key areas), forming "one belt, three cores, ten districts linked, blue-green slow-mobility ring"; Block Token zone licensing is the core of the urban AI governance protocol.

Tier	Area	Scope & Task
Study Scope	43.6 km^2	Haidian southern innovation corridor: N. 5th Ring Rd north, Jingzang Hwy east, Xizhimen Outer St south, Wanquanhe Rd west; global AI ecology, strategic positioning & future urban form
Overall Design Scope	11.41 km^2	Urban districts & industrial areas within 1—2 km of the heritage park; overall renewal framework, industrial space, blue-green slow mobility, transport-utilities & urban-form controls
Key Areas Scope	368.4 ha	Zhongzhiyuan 192.1 + Origin 104.3 + Dazhongsi 72.0; refined design of three core districts north-to-south

6 04 Study: Global Cases & Regional Synergy

Transferable lessons from 6 global AI innovation benchmarks

Lessons from Cases

- Mission Bay (San Francisco, USA): capital token + R&D token dual-track → Origin Community
- Silicon Roundabout (London, UK): brownfield renewal logic → Dazhongsi
- MaRS Discovery District (Toronto, Canada): tripartite governance model → Zhongzhiyuan
- Cornell Tech (New York, USA): campus-industry sandbox → Origin conversion street

"Three districts, two wings" synergy

"Three districts, two wings" synergy: Zhongguancun tech-service wing (capital + IP), Xiaoyuehe scenario wing (Xueyuan Rd linkage), Future Science City/Huairou (basic-research spillover), E-Town (hardware translation).

Case	City	Scale	Key Mechanisms	Lesson for JZ
Mission Bay	San Francisco, USA	1.23 km^2	VC cluster + campus-industry-research fusion	Capital token + R&D token dual-track → Origin Community
Silicon Roundabout	London, UK	≈4 km^2	Brownfield renewal × creative industries	Brownfield renewal logic → Dazhongsi
MaRS Discovery District	Toronto, Canada	139,000 m^2	Innovation hub + centralised research sharing	Tripartite governance model → Zhongzhiyuan
Cornell Tech	New York, USA	12 acres	Open campus × corporate co-built sandbox	Campus-industry sandbox → Origin conversion street
Kendall Sq.	Boston, USA	≈2.6 km^2	Very high walkability × dense innovation network	Walkability grading → public space
Punggol	Singapore	50 ha	System-level interconnection + microgrid	System interconnection (not building from scratch)

7 05 Overall Design: Land Use & Renewal

Four functional land uses cover 11.41 km^2, no overlap, no gaps

Land-Use & Renewal Strategy

Four functional land uses total 1141.3 ha (=11.41 km^2), full coverage, no overlap: R&D innovation 311.9 ha (27.3%), green open space 270.8 ha (23.7%), industry & commerce 309.4 ha (27.1%), talent community 249.2 ha (21.8%). Renewal uses the demolish/convert/retain concept draft; all claims confidence=low, pending building census.

Land-use calibre note

Four land uses total 1141.3 ha = 11.41 km^2; green ratio counts all green patches (incl. pocket parks & waterfront).

Land Use	Area	Share
LU-001	311.9 ha	27.3%
LU-002	270.8 ha	23.7%
LU-003	309.4 ha	27.1%
LU-004	249.2 ha	21.8%

8 06 Transport, Rail & Utilities

TOD stitching + continuous slow mobility + quantified scenario metrics

Mobility Metrics Detail

Indicator	Value	Calibre
TOD 500m Union	20.6%	Union of 500m walk circles at Wudaokou/Qinghua E.Rd W/Dazhongsi (EPSG:4548)
Green 300m Coverage	85.6%	300m walking coverage of green patches (EPSG:4548 measured)
Greenway Gaps	15	Existing gaps on the park spine (measured); stitching improves connectivity
Building Density	TBD	FootprintTBD tal 11.41 km^2, pending official census

Transport System

TOD integration: stitching Wudaokou/Qinghua E.Rd W/Dazhongsi stations, exploring 15-minute rail-walk circles; elevated corridor between Dazhongsi & Wudaokou under study (pending municipal feasibility)

Slow mobility & green transport: continuous cycling & walking spine along the heritage park; autonomous shuttle micro-loop concept route

Quantified scenarios (synthetic): green 300m coverage 85.6%; three-station 500m union 20.6%; greenway spine gaps ≈15 (EPSG:4548 measured)

Utilities & energy: rooftop PV + microgrid exploration; concealed edge-compute micro-stations at park nodes (conceptual); data minimisation & 7-day deletion cycle

9 07 Blue–Green Space & Cityscape

One spine, two rivers, multiple corridors, hundred parks: green, wind corridors & Qinghe low–carbon waterfront

Blue–Green Metrics Detail

Indicator	Value	Breakdown
Green Ratio	31.1%	10 green patches totalling 3.55 km^2 (total 11.41 km^2); LU–002 central parks 270.8 ha per land–use classification
Green–Spine Wind Corridor	9.5 km	Wind corridor along the park green spine; estimated cooling 0.8—1.5°C (synthetic, pending CFD)
Blue–Green Pattern	One spine, two rivers, corridors & parks	Park spine + Qinghe/Xiaoyue rivers + greenways linking 12 key communities & campuses

Blue–Green Network

• One spine, two rivers, corridors & parks: heritage park as north–south ecological spine with Qinghe & Xiaoyue rivers; continuous greenways link 12 key communities & campuses

• Green system (measured): 10 green patches totalling 3.55 km^2 (31.1%); LU–002 central parks 270.8 ha per land–use classification

• Wind corridor & microclimate: 9.5km green–spine dominant corridor, estimated cooling 0.8°C—1.5°C (synthetic range, pending micro–weather stations & CFD)

• Qinghe low–carbon waterfront: multi–modal ecological sensing (water quality / bird song / microclimate); smart stormwater & carbon–sink regulation (concept)

Material	Expression
Centennial JZ Industrial Red Brick	Heritage · along Heritage Park
Zhongguancun Tech Grey Aluminium	Modern industry · park interface
Future AI Transparent Glass	Futuristic · landmark nodes

10 08 Detailed Design of Three Key Areas
Three "stations": Departure Yard / Zero–Km Station / Marshalling Yard

Key Area Design Points

- Zhongzhiyuan AI Acceleration Area (192.1 ha · token: Departure Yard): garden–style full–stack innovation district: Qinghe waterfront restoration, national AI model test ground, hardware/software compatibility labs; anchors: Safety Sandbox · Qinghe Low–Carbon Innovation Waterfront
- Beijing AI Origin Community (104.3 ha · token: Zero–Km Station): campus–adjacent commercialization community: stitching Qinghua E.Rd W/Wudaokou gaps, converting legacy plants into low–cost open–source co–working; anchors: Open–Source Launch Hall · Conversion Street
- Dazhongsi AI Industry Cluster (72.0 ha · token: Marshalling Yard): urban smart–economy district: transit TOD integration, four–quadrant elevated walkway linking commercial & industry HQs; anchors: Intl Roadshow Hall · Data–Asset Hall

Key Area Design Points

Zhongzhiyuan AI Acceleration Area (192.1 ha): garden–style full–stack innovation district: Qinghe waterfront restoration, national AI model test ground, hardware/software compatibility labs

Beijing AI Origin Community (104.3 ha): campus–adjacent commercialization community: stitching Qinghua E.Rd W/Wudaokou gaps, converting legacy plants into low–cost open–source co–working

Dazhongsi AI Industry Cluster (72.0 ha): urban smart–economy district: transit TOD integration, four–quadrant elevated walkway linking commercial & industry HQs

Key Areas	Area	Block–Token Role
Zhongzhiyuan AI Acceleration Area	192.1 ha	Departure Yard

Key Areas	Area	Block-Token Role
Beijing AI Origin Community	104.3 ha	Zero-Km Station
Dazhongsi AI Industry Cluster	72.0 ha	Marshalling Yard

11 09 AI Innovation Ecosystem & 13 Scenario Cards

13 scenario cards + 6 personas + 3 AI landmarks

Scenario Access Distribution

13 scenario cards graded by spatial access: L1 open display 8 (TRL 5—9), L2 conditional deployment 4 (TRL 4—7), L3 governance sandbox 1 (TRL 6—7). Five-state verification covers all 13 cards (synthetic).

6 personas

- Open-source developers: 24h launch hall & co-working space
- Startups: shared edge-compute stations & model red-team sandbox
- Corporate visitors: intl roadshow hall & TOD express corridor
- Residents: slow-mobility green loop & embedded community service points
- Students & faculty: campus-park stitch paths & commercialization stations
- International visitors & academic guests: interpretation lounge & academic release centre

01 Open–Source Launch Hall Origin Community L1 · TRL 7–8	02 Safety Governance Sandbox Zhongzhiyuan L3 · TRL 6–7	03 Edge–Compute Station All–line nodes L2 · TRL 6–7	04 AI Slow–Mobility Wayfinding Heritage Park spine L1 · TRL 5–7
05 Dazhongsi Intl Roadshow Hall Dazhongsi L1 · TRL 8–9	06 Qinghe Low–Carbon Innovation Waterfront Zhongzhiyuan · Qinghe frontage L2 · TRL 4–6	07 Campus–Adjacent Conversion Street Origin Community · campus edge L1 · TRL N/A	08 Data–Asset Hall Dazhongsi L2 · TRL 5–6
09 AI Lifestyle Demo Street Community / commercial hub L1 · TRL N/A	10 Global AI Week Route Full–line blue–green space L1 · TRL N/A	11 Multi–Species Sensing Node Zhongzhiyuan · Qinghe frontage L2 · TRL 4–5	12 Analog Alternative Station Origin / Dazhongsi L1 · TRL N/A
13 Time–Fairness & Zone–Sharing Card Along Heritage Park L1 · TRL N/A			

3 AI Landmarks

- Jing–Zhang AI Meridian Tower (Park × Qinghua E.Rd)
- Open–Source Light Launch Hall (Origin Community)
- Qinghuayuan AI Origin Memory Park

12 10 Block-Token Governance & Open Operation

L1-L3 spatial access × five-state verification lifecycle, orthogonal

Five-State Lifecycle

- Proposal: unverified claim
- Desk-sim passed: synthetic simulation passed
- Pilot pending: awaiting on-site authorisation
- Authorised operation: operating with token
- Hard-stop / return: halt & rollback audit

Governance Protocol

Civic Value Protocol: proposes studying a 10%—20% share of commercial edge-compute nodes and AI scenario test revenue along the corridor into a "Jing-Zhang Community Public Value Fund" (voluntary CSR charter under Zhongguancun sandbox pilot framework; final ratio & rules per competent authorities and community representatives).

Civic Value Protocol + City-as-Repo

Civic Value Protocol: proposes studying a 10%—20% share of commercial edge-compute nodes and AI scenario test revenue along the corridor into a "Jing-Zhang Community Public Value Fund" (voluntary CSR charter under Zhongguancun sandbox pilot framework; final ratio & rules per competent authorities and community representatives).

City-as-Repo: spatial Pull Request (GeoJSON impact area + compute power + noise assessment) → tripartite Code Review (governance committee / compliance / community reps) → one-click rollback (24h shutdown & revert to spatial baseline).

Access Level	Zone	Gate & Safety
L1 Open Display	Dazhongsi Roadshow Hall, etc.	Filing-based; mature tech displayed directly; data de-identified & public; disconnect immediately on risk
L2 Restricted Testing	Origin Community Conversion Street, etc.	Joint approval + ethics review + human oversight; strict access control, human takeover on anomaly
L3 Sandbox Validation	Zhongzhiyuan Test Ground	Heavy supervision + whitelist + expert review; physical isolation, hardware kill-switch, forced data destruction

13 11 Renewal Project List & Phased Delivery
All six renewal projects in the Block-Token Proof-Mile loop

Phased Implementation

Six renewal projects phased as near-term (3), mid-term (2) and long-term (1); each runs the Block-Token Proof-Mile loop: small-scale verification (metrics met) before scale-up; failed verification triggers risk hedging.

Renewal Project List

Project	Name	Phase	KPI Formula	Risk
JZ-01	Slow-mobility gap stitching	Near-term	Connectivity = connected gaps / total gaps	Funding gap → switch to ground guidance
JZ-02	Qinghe ecological experience concept	Near-term	Green survival rate > 90%	Flood risk → lower retrofit standard
JZ-03	Factory retrofit & conversion	Mid-term	Occupancy rate > 80%	Weak leasing → convert to general office
JZ-04	Corridor system feasibility	Mid-term	Corridor daily flow > baseline	Structural limits → drop elevated corridor
JZ-05	Compute-centre scenario study	Long-term	Measured PUE < 1.2	Energy overrun → throttle & downgrade
JZ-06	Smart component deployment concept	Near-term	Device online rate > 95%	Privacy dispute → shut down sensing modules

14 12 Indicator System & Compliance Matrix

All metrics traceable to metrics.json; standards covered in standard_matrix.json

Metric Calibre

All metrics traceable to metrics.json; spatial metrics (green 300m coverage, TOD 500m union, greenway gaps) use EPSG:4548 measured calibre; conceptual estimates marked synthetic, recalculated after official regulatory plan release.

Metrics & Standards

Standard / Basis	Coverage	Status
PROJECT–OFFICIAL–ANNOUNCEMENT	Announcement clauses 1.3/1.4/1.5 coverage	Responded
PROJECT–AGENT–OPEN–CALL–TASKBOOK	Three positions / five functions / agent.1–6	Responded
MOHURD–URBAN–DESIGN–MEASURES	Urban design methods & management requirements	Responded
MOHURD–CONTROL–DETAILED–PLANNING	Regulatory–depth urban design organisation	Responded
MNR–LAND–USE–CLASSIFICATION–GUIDE	National land–use classes 0802/05/1401/0702	Responded
MOHURD–ARCH–DESIGN–DEPTH–2016	Architectural design depth framework	Responded
GENERATIVE–AI–INTERIM–MEASURES	Generative AI service management boundary	Responded
BARRIER–FREE–ENVIRONMENT–LAW	Barrier–free environment law (Art. 39 boundary)	Responded
ELDERLY–SMART–TECH–PLAN–2020–45	Elderly services & smart–tech parallel policy reference	Responded

15 13 Risk, Copyright & Compliance

Risks, copyright & compliance: conceptual proposals are not approval conclusions

Versioning & Rollback

City-as-Repo: spatial Pull Request (GeoJSON impact area + compute power + noise assessment) → tripartite Code Review (governance committee / compliance / community reps) → one-click rollback (24h shutdown & revert to spatial baseline).

Compliance Statement

• Provisional boundary: official polygons not released — all boundaries/metrics provisional (official_boundary=false), concept generation & display only; full recalculation upon official data

• Simulation & pilot dual-track: weather simulation / crowd simulation / energy estimates are Synthetic Tabletop; pilots require desk-sim sign-off + authority authorisation + 5-step rollback sequence

• Copyright & legal: COMMUNITY-DISPLAY-ONLY licence; conceptual proposals are not administrative approval conclusions; no unauthorised portraits/logos/copyrighted materials

• Data gaps: building census / regulatory FAR / real road network unavailable — demolish/convert/retain classes & FAR pending official data

16 14 Sheet Index & References

Sheet index & references: display layer defers to structured data

Structured Data Flow

This booklet is the display layer; all conclusions defer to the structured data layer: GeoJSON (boundary/green/buildings) → metrics.json (metrics) → compliance_matrix.json (standards) → this booklet's figures. Any conflict is resolved by structured data.

Sheet Index & References

Source list: brief/public-brief.md (primary authority), data/source_registry.json (registry), data/processed/agent_fact_pack.md (navigation layer).

Sheets / Deliverables	Notes
site-overview.png	Overall location (provisional)
land-use-structure.png	Land-use structure (provisional)
key-areas.png	Key areas (provisional)
proposal.md	Master (ref)